

Figure 2: (a) An NAND gate driving a large capacitance. (b) A super buffer is inserted.

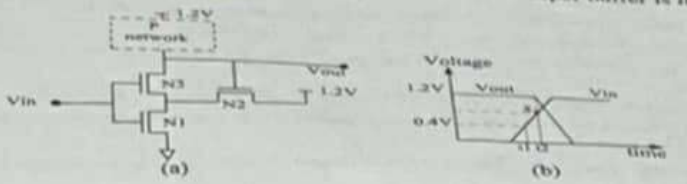


Figure 3: (a) A CMOS logic gate (b) waveforms of Vin and Vout.

voltage of the output Vout is 1.2V and an input stimulus (0→1.2V) as shown in Fig. 3(b) is applied, how much voltage of the input stimulus (Vin) at x is? (4 points) Identify the operation region of the transistors N1, N2, and N3 from t1 to t2. (6 points)

5. (10 points) Fig. 4(a) shows a CMOS circuit. Assume that the sizes of NMOS and PMOS transistors in the logic gates are W/L and 2W/L, respectively. Also, $\mu_n = 2\mu_p$. If the functional inputs of the circuits are (ABCD)={0011, 0101, 1101, 1110}, explain the relationship of noise margins NM_L and NM_H (i.e., $NM_L > NM_H$, $NM_L < NM_H$, or $NM_L = NM_H$) of each gate.

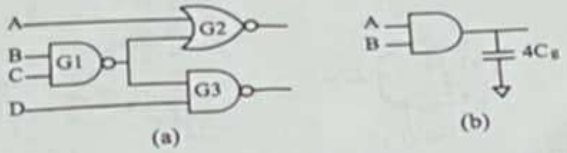


Figure 4: CMOS circuits

6. (10 points) For a given technology, the turn-on resistance of a W/L NMOS is R and $\mu_n = 2\mu_p$. Also, the gate capacitance of a W/L NMOS or PMOS transistor is C_g. Fig. 4(b) shows a two-input AND gate drives a 4 C_g capacitive load. Design the AND gate such that the worst-case delay of the AND gate is smaller than $8RC_g$, where the delay is estimated by using simple RC model and the parasitic capacitance can be ignored. Draw the transistor-level schematic diagram and label the size of each transistor in terms of W/L.

7. (15 points) Realize the following Boolean functions using CMOS logic. (a) $F = X(Y + Z) + XW$ (5 points) (b) $F = \overline{X(Y + Z)}$ (5 points) (c) $F = XY + YZ$ (5 points).